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Master Thesis in Natural Computation

A lognormal based framework for human motion analysis - generation in 3D spaces

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Then what is your hand for?

Rei Ayanami, End of Evangelion

To effectively contain a civilization's development and disarm it across such a long span of time, there is only one way: kill its science.

Liu Cixin, The Three-Body Problem

If one's different, one's bound to be lonely.

Aldous Huxley, Brave New World

The saddest aspect of life right now is that science gathers knowledge faster than society gathers wisdom.

Isaac Asimov, Book of Science and Nature Questions

אמת

The Golem of Prague

The true measure of a man is not his intelligence or how high he rises in this freak establishment. No, the true measure of a man is this: how quickly can he respond to the needs of others and how much of himself he can give.

Philip K. Dick, Our Friends

Acknowledgements

Write acknowledgements here.

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1 Introduction

Initially it was through the steam engine, then the electricity, then robots, and now it is through their collaboration with humans that humankind has been able to improve productivity in workplaces. The times when robots were used to replace humans in caged hazardous environments to perform repetitive tasks are now behind this day.

The human wish to bring to life a being in their own image has been a long-standing dream, often expressed in literature and cinema. From the Golem of Prague to the replicants of Blade Runner, from their first appearance Rossum's Universal Robots to the robots of Isaac Asimov's stories.

In the following two sections, the problem will, firstly, be defined and, secondly, its relevance in the field of computer engineering will be discussed. The goal is to provide a clear understanding of the problem at hand, to justify the need for a solution and to highlight the potential impact of the proposed solution on the field of computer engineering.

Once the problem has been sufficiently explained, the subsequent chapters will provide the reader with an overview of the current state of the art, the design behind the proposed solutions, an analysis on the results found, and finally a conclusion on the work done, highlighting the contributions made to the field of computer engineering and the potential for future research in this area.

1.1 Problem Definition

Nowadays, robots collaborate with humans in a shared workspace, where they assist the workers in performing variable tasks that require precision. This collaboration is made possible by the development of advanced sensors, actuators, and control algorithms that allow robots to perceive and respond to the environment in real-time.

For this reason, modern robotics — collaborative robotics particularly — has become an interdisciplinary field. Teams of engineers with different backgrounds, such as mechanical, electrical, computer and software work together to design and develop robots that can collaborate with humans in a safe and efficient manner.

On this thread, the following work addresses the problem of safety and efficiency in human-robot interaction. One of the most underestimated aspects of this problem is the human percep-

tion of the robot's motion. To be more predictable, it has to be human-like, since we as humans perceive something to be "*better*" when it resembles us. To this end, the solution regards endowing robots with the skill of moving like humans.

Human-like motion increases predictability, safety and comfort during the robot interaction with a human. As workers are used to cooperate with other humans, they expect the robot to move similarly. If the robot moves through unexpected fast uniform movements, the human partner may be startled, or the robot may hurt the human partner. On the other hand, if the robot moves as a human worker would, the human partner can predict its movements. By predicting the robot's movements, the human partner can adjust their own movements accordingly, reducing the risk of collisions and improving overall safety.

Thanks to the predictability of the robot's movements, the human partner can also feel more comfortable and confident in interacting with the robot. This can lead to increased productivity and efficiency in collaborative tasks.

To teach this skill, its characteristics must be captured in a set of features learnable by a generative model. These features should explain the underlying principles of human motion, and be applicable to any dataset of human movements. Once a model learns these features, it can generate new movements without overfitting a large set of parameters over a specific dataset, which would make the model less generalizable and more complex.

A generative model able to generalize means reducing the amount of data needed to train it and also avoid the need for fine-tuning. This is a crucial aspect, as the motion must be generated during the robot's operation, i.e. in real-time. This requirement means that, if the model does have too many weights, the robot must mount highly performant hardware, thus inherently increasing the cost of implementation.

Avoiding fine-tuning needs is mandatory, as not all situations can be expressed inside a dataset. When a human worker has to solve a task, they do not observe their dataset of previous actions to proceed. Instead, they develop a meaningful representation of their past actions in order to understand the solution. Thus, endowing the same methodological reasoning inside this work is inevitably necessary, otherwise those desired results will never be achieved, and the generative model won't be able to operate in any environment, as no task is the same as the previous ones.

1.2 Problem Relevance in Computer Engineering

2 State of the Art

The following section discusses the current advancements in human-robot interaction, and the limitations of the current literature. The goal is to provide the reader with a comprehensive overview of the state of the art in this field, highlighting challenges and opportunities for future research.

2.1 Current Research in Human-Robot Interaction

The interaction between humans and robots is a complex problem that spans various fields: robotics, human-computer interaction, psychology, neuroscience, and so on. Humans perceive robot motions comfortable when resembling human-like movements, quantifying this aspect as the HL (Human Likeness) index [1]. But in the scenario of interaction.

2.2 Literature Limitations and Open Challenges

3 Proposed Solution

This chapter gives a comprehensive overview of the solution adopted to address the problem at hand, starting from the device chosen, going through the data collection and processing systems, then ending with the generative model and its performance evaluation. The goal is to provide the reader with a clear understanding of the choices, weighing pros and cons, so to justify the decisions made.

3.1 Methodology

As required by the problem to solve, the chosen methodological approach focuses on the data. This decision was natural, as having to put together various independent system means looking for a common interface for all of them. And in this case, that common interface is the data flow.

This data is the very essence of the work, and its flow and change must be carefully observed. From the data acquisition, human motion characteristics must be present in its content. There may be errors from the collection device (measurement), errors from the collection itself (bias), errors from the software modules themselves (bugs), and finally, error from the generative model itself, due to a wrong architecture.

Having in mind this flow, the methodology originates immediately, and asks for modularity. Someday, the data collection may be done through motion trackers, recording the motion of another part of the body. Meanwhile, instead of choosing one step adopted by this solution, someone else may try a different technique. Whatever is the reason behind the change, and the module changed, the flow of data and its meaning will always be the same.

Other than modularity, in this methodology the data flows through an ordered sequence of actions: a pipeline. While this aspect seems to regard more the system design, it is important to have in mind that also the decision of the collection method, the device it needs, the task assigned, and the experimental campaign themselves are steps that shape the data during its flow. And these decision may be changed later, but the pipeline must still work nonetheless.

3.2 System Design

The requirements that arose from choosing this methodology now apply to the composition of subsystem inside the overall system itself. Each subsystem corresponds to an interchangeable module; as required by the methodology adopted each is a step of the pipeline in charge of managing data.

The three main subsystems are the following

- The data collection system concerns recording the human motion data
  - The experimental campaign, containing experimental protocol and designed tasks
  - The data collection hardware and software
  - The kind of data expected in input
  - The collected dataset provided as output
  - The interface with the user, which contains a set of tasks that must be formally defined to be also interpretable by a model
- The data processing system is responsible for extracting the features that best describe the recorded data
  - The interface with the data collection system, through accessing the collected dataset
  - The codification of a theoretical model behind the reconstruction of the collected data through optimization problems
  - The reconstructed dataset, which entries are the reformulation of the collected data through a set of feature, explained by the model adopted
- The data generation system manages the generation of new human motion data
  - The interface with the data processing system, through reading the reconstructed dataset
  - The interface with the tasks, which interpretation must be the same of the human subject to not introduce bias in the dataset
  - The neural network model architecture, its training and generation methods
  - The training procedure defined by the loss function, the set of features to take as input, as recurrent input from the previous task, the expected outputs, and the data sets

The following subsections provide a deeper insight into each system, without focusing on the implementation of them, which main modules are instead detailed in the next three sections,

one for each subsystem. This more general overview serves to understand how the three system work in an architectural point of view.

Regarding the reasoning behind parameters or techniques adopted by these subsystems, their explanation is instead given in the next chapter.

3.2.1 Data Collection System

The data collection system is responsible for recording the human motion data. It is composed by the experiments themselves, the haptic device used to record the human motion, and the software that manages the communication between the experimenter, the haptic device and the human subject.

Haptic Device

The device adopted for the problem is *Touch*, from *3D Systems*. It is a haptic device with six degrees of freedom, allowing it to move in three-dimensional space and rotate around three axes. It is equipped with a stylus that can be held by the user, allowing them to interact with the device. The device is able to produce some linear forces, which are applied to the user's hand through the stylus.

This kind of devices is commonly used in teleoperation, in which the user controls a remote robot by moving the stylus and feeling the forces applied by the robot. In this case the device is used only to collect data from the user's movements.

3.2.2 Data Processing System

The data processing system is responsible for processing the collected data by extracting the features that describe it according to the theoretical model. This model is based on the theory of kinematics of rapid human movements, which describes human motion as a sequence of primitives, called lognormal strokes, that can be superimposed to generate complex movements.

3.2.3 Data Generation System

The data generation system is responsible for generating new human motion data, based on the features extracted from the collected data, and the task to perform described in an experiment.

3.3 Haptic Interface Software

The haptic interface software manages both the user interface of the haptic device and the communication with the robot.

From the side of the experimenter, the software component allows them to control the experiment execution, by selecting its configurations, like the path to show, the number of repetitions, the sampling frequency, and so on.

From the side of the human subject, the software component interfaces with them through a simulated environment that projects the position of the haptic device inside a virtual space. In this space, the path is represented as an ordered sequence of spheres, while the device is represented as a sphere that can be moved by the user through their interaction with the device.

3.3.1 `ROS2` Controller

The haptic device controller is in responsible for managing the communication between experiment, haptic device and subject through sending commands and receiving data, and of the experiment execution.

The controller takes as input the configuration of the experiment, which includes the path to show, the number of repetitions, the size of the waypoints, the forces to apply and so on. From this information, during the execution it is able to communicate with the experiment through the keyboard, and with the haptic device through its command and state interfaces. The controller is also responsible for collecting the data from the haptic device, which includes the position, orientation, linear and angular velocities.

Furthermore, it is in charge of communicating with the subject through the screen via the visualization of the path during the pre-training phase. In this case, the role of the controller is to communicate a series of colored waypoints, and the current position of the haptic device to the visualization software `RViz2`, which is responsible for rendering the scene on the screen. The controller applies the haptic feedback to the user, by computing the forces to apply to the haptic device based on the current position of the device and the position of the next waypoint, multiplied of a gain factor and the distance between the two.

3.4 Data Processing Pipeline

In the data processing pipeline the collected data is transformed into a representation that reflects the adoption of the theoretical model of human motion. This is done by finding a set of features that, when combined, can reconstruct the original data sufficiently well. In order to do so, a formal definition of the theoretical model is required, as its implementation is given in these software components.

3.5 Generative Model

The generative model is trained with the dataset of movements, extended with the temporal and spatial characterizations, to learn the underlying principles of human motion. Once trained, the model can generate new movements that are similar to the original ones, by providing the path in the space and its time characterization through a velocity profile derived from the lognormal parameters. The generated movements can then be executed on the same system that collected the original data.

4 Experimental Validation

4.1 Task Design

The goal of the experiment campaign is relatively simple: the collection of a dataset of rapid and natural human movements. However, in order to obtain a dataset of sort, it is mandatory to provide the same task to each subject. Nonetheless, the same input must be grasped equally by different subjects. Thus, the deletion of ambiguity in task definition is imperative.

4.1.1 Waypoint Definition

As previously mentioned, a path is defined as an ordered sequence of points in the space, called waypoints, each defined by its position in the space. Visually, they are represented through spheres, which size is a trade-off between the accuracy of the collected data and the naturalness of the user's movements. However, even though the user should try to follow the path by crossing the waypoints, some deviation from them is to be expected, since no human movement ever perfectly follows a path. In the same way, a generative model should think of waypoints as a reference, akin to an attractor, rather than a strict constraint.

A path of two waypoints is simply a straight line between the two. On the other hand, a path of three waypoints is interpretable as a curve, since the user is supposed to follow the path by trying to reach the waypoints, but not strongly forcing themselves to cross and step on them: the result is a curve that starts from the first waypoint, gets near the second one, and ends almost on the third one. The same principle applies to paths of more than three waypoints, which are interpreted as a sequence of curves, which curvature is inversely related to the alignment of the waypoints.

The issue with such a representation is that it doesn't provide any information about the curvature of the path, neither about the depth of the waypoints in the space.

4.1.2 Path Description

A path, in simple terms, is an ordered sequence of waypoints. This description would be sufficient for a model, since its perception of a three-dimensional space is mathematical and thus

unmistakable. However, a human's perception of a three-dimensional space is not mathematical, and thus it is not as accurate as the model one. Therefore, it is necessary to provide the human subject with a more intuitive representation of the path.

A path could represent a common task that a human subject may perform in their daily life, such as moving an object from one place to another, or using a particular tool. The path should be designed to be as intuitive and natural as possible, eventually combining the geometrical description with a semantic one, ensuring that the subject grasps and performs it smoothly, avoiding unnatural movements that could poison the collected data.

Visual & Haptic Feedback An intuitive representation should make use of the human natural instinct to merge different sensory inputs to form a more complete perception of the environment. In this case the two only classes of sensory inputs available are the visual and haptic ones, since they are the only two that are directly transformed into positional information in the perceived space by the motor cortex.

The visual feedback is shown to the subject through the ordered sequence of spheres projected in a simulated three-dimensional environment in the screen. The depth perception is given by the perspective of the camera, which can be moved by the subject to change their point of view.

The haptic feedback is given to the subject through the haptic device's actuators, which apply forces to the subject's hand to guide them towards the path. These forces are applied toward the next waypoint, and let the user stay on each waypoint to give them time to memorize the positions of the waypoints. The more the user's ability to follow the path increases, the less the haptic feedback is applied. More importantly, the rest time on the waypoints is zeroed, since the user must follow the path without stopping on the waypoints. This should happen some time before the haptic feedback is removed, in order to let the user get used to the fact that they are supposed to follow the path without stopping on the waypoints.

Motor Representation Both the haptic feedback and the visual feedback are not used during the recording phase, but rather in a previous phase called pre-training. During this step, the user is aided in following the path through the visual and haptic feedback, until a certain level of proficiency is reached. This allows the subject to obtain a representation of the path inside their motor cortex, which is a fundamental aspect of human motion. Once this representation is formed, the user starts to follow the path without any aid, in order to test this proficiency, the feedbacks are slowly removed, until they are completely disabled. The user is then asked to follow the path without any aid, while the system collects their movements.

Curvature Representation The only issue with representing a path only by a sequence of points is that it only takes into account direct links between waypoints. Humans rapid movements don't go straight from one point to the following one. Instead, they try to store momentum

by not stopping on each point, but making curves during the execution. This aspect must be extracted from the dataset and learned by the generative model, since it is a fundamental aspect of human motion.

An idea to make this aspect stand out from the collected data could be the introduction of arrows in the visual environment, although this means crowding it with more information that could be distracting or hard to interpret for the user. Also, this requires defining other tunable parameters, since the arrows would have to be placed at a specific distance from the waypoint it starts from, with a specific length, at a specific size and orientation. However, by only specifying the waypoints, the user is free to interpret the path in their own way, and thus they can execute it in a more natural way, which is what the experiments aim to achieve. In the end, it is mandatory to explain to the user that they are supposed to follow the path by trying to reach the waypoints, but not strongly forcing themselves to cross and step on them, since this would lead to unnatural movements: the user should be encouraged to follow the path smoothly.

4.1.3 Path Taxonomy

The paths that will be shown to the subject during the experiment are grouped into the following categories:

- Burst: impulsive movements. Simple paths that help the subject getting used to the haptic device.
  - Straight line over the XY, YZ or XZ planes
  - Semicircumference over the XY, YZ or XZ planes
- Geometric: two- and three-dimensional round geometric shapes
  - Helix, growing over the X or Y axes
  - Spiral, in the XY plane
  - Ellipses, one in the XY plane and the other in the XZ plane
  - Circle, one in the XY plane and the other in the XZ plane
- Symbol: letters of an alphabet
  - Cursive lowercase L (ℓ) in both the XY and XZ planes
  - Print uppercase S in an oblique plane in both senses
  - Cursive Greek Phi (φ) in the XY plane
  - Print uppercase N which segments differ in plane

4.2 Evaluation Metrics

4.3 Experimental Protocol

The real challenge is to design the experiments in such a way that there isn't difference between the information received by the subjects during the experiment and the one received by the generative model during its training phase.

4.3.1 Sources of Bias

To ensure consistency across different subjects, the experimental protocol should be designed to minimize any potential biases or confounding factors that could affect the results. For instance, it is necessary to simplify some complex terms by finding more intuitive and user-friendly ways to explain them, so to ensure user comprehension and engagement.

Additionally, the protocol should be designed to minimize any potential sources of error or variability in the data collection process, such as differences in the user's physical abilities or environmental factors that could affect their movements. This last point is particularly important, since the experiment is designed to be executed in a laboratory environment, where the user is asked to sit on a chair and interact with the system through a haptic device. The laboratory environment should minimize any potential distractions or sources of noise that could affect the user's movements and the data collection process.

Posture The environment must be designed to ensure the user's safety and comfort, by providing them with a comfortable chair and ensuring that the haptic device is properly calibrated and adjusted to their physical characteristics (i.e. dominant hand, arm length, height, etc.) to reduce any potential sources of discomfort or fatigue and increase comfort during the experiment. The more the user feels relaxed, the more natural their movements will be, and thus the more accurate the collected data will be.

Observer effect The user is aware of the fact that they are being observed and that their movements are being recorded, which could affect their behavior and introduce some bias in the collected data. To prevent this kind of bias, the user should be informed about the purpose of the experiment and the data collection process, and they should be given the opportunity to ask questions and provide feedback. This will help to ensure that the user feels comfortable and engaged in the experiment, and that they are motivated to perform their best. Of course, any reveal of the purpose of the experiment should be done in a way that doesn't influence the user's behavior, and thus bias the collected data.

4.3.2 Tunable Parameters

The real challenge is to design the experiments in such a way that there isn't difference between the information received by the subjects during the experiment and the one received by the generative model during its training phase.

Parameters The session does have various parameters that need to be tuned in a trade-off between the accuracy of the collected data, their naturalness, the fatigue of the user, the length of the experiment and the amount of data collected. The parameters that can be tuned are:

- W_s: The size of the spheres that represent the waypoints of the paths, which can be increased to make it easier for the user to follow the path, but at the cost of reducing the accuracy of the collected data. A bigger sphere would require less training and less precision from the user, but it would also make the collected data less accurate, since the user would be able to deviate more from the path without being penalized. On the other hand, a smaller sphere would require more training and more precision from the user, but it would also make the collected data more accurate, since the user would be forced to follow the path more closely. Although, this forceful guidance could lead to unnatural movements, unnatural to a human, which would be detrimental to the generative model.

- P_l: The number of waypoints of a path, which can be increased to make the path more complex, at the cost of increasing the difficulty for the user to follow the path. A more complex path would require more training and heavier cognitive effort, while increasing the data collected. Conversely, a simpler path would require less training and less cognitive effort, while reducing the data collected. The complexity of the path needs to be balanced with the user's ability to follow it, to collect accurate and natural data.

- P_r: The number of repetitions for each path, which can be increased to collect more data, at the cost of increasing the fatigue of the user and the length of the experiment. This parameter comes at the cost of making brief rest periods between repetitions, which would compensate the user's fatigue, but at the cost of greatly increasing the time duration of the full experiment session.

- T_r: The number of execution prior to the recording phase, named "pre-training phase". Albeit it would be simpler to have a fixed number of repetitions, the issue is within the variability of human learning. Some subjects may learn a path in less time than other, thanks to a prior knowledge of environments over the screen, or of their . Since this parameter depends on the user's ability to follow the path, and on the complexity of the path itself, it is left to decide up to the experimenter. If the user correctly follows the path for a certain number of repetitions, and feels enough confident to proceed to the recording phase, they can switch to the recording phase. Until then, the pre-training will aid the participant by providing the haptic feedback.

- N_P: The number of paths, which can be increased to collect more diverse data, at the cost of increasing the length of the experiment. More paths would provide a better representation of the task space and allow to more generalization, but would also require more time and effort from the user. Although, having a few paths would limit the diversity of the data collected, and thus the generalization of the generative model.

- R_t and R_f: The rest periods length and frequency (expressed as a fraction of N_P), which can be increased to reduce the fatigue of the user, at the cost of increasing the duration of the experiment. The rest periods need to be short to keep the user focused on the task, but long enough to allow them to recover from their fatigue, while not being felt coercive by the user. Moreover, the rest periods need to be frequent enough to allow the user to recover from their fatigue while executing the different paths, but not too frequent to make the user lose focus on the task.

4.3.3 Parameter Values

Regarding the aforementioned experimental parameters, the following values have been chosen:

Parameter	Description	Value
W_s	Waypoint diameter	0.05 m
P_l	Path length	2-7 waypoints
P_r	Path repetitions	20 repetitions
N_P	Number of paths	20 paths
R_t	Rest time	60s
R_f	Rest frequency	$1/6$

As explained before, the training repetitions are not a tunable parameter, since they depend on the subject ability to follow the path, and on the complexity of the path itself. At most, it would be possible to define an acceptance criterion during the training phase. Since the training requires them to cross all the waypoints, it is safe to assume that the criterion is in some way already encoded in the waypoint size W_s.

4.3.4 Written Protocol

Providing the user with a written protocol is mandatory not only to ensure that all the user have the same information, but also to ensure that they have the same understanding of the task. From the different equivalent sequences of words available to explain an instruction, the most intuitive should be chosen, to avoid any bias in the collected data.

Input In each experiment of a session a path is shown to the user, their goal is to traverse close to the waypoint as fast as possible, with the understanding that it's not strictly necessary to cross all the waypoints, but rather to think of them as attractors. They will first be pre-trained

to follow the path with the aid of visual and haptic feedback, until they reach a certain level of proficiency, defined by the experimenter, and then they will be asked to follow the path without any aid, while the system collects their movements.

Result For each execution of a path, the system collects a sequence of timestamped points in the space, which are expressed in Cartesian coordinates with respect to the reference frame. Furthermore, the tangential velocity of the user's hand is computed at each timestamp. By interpolating these points, and by having a high sampling frequency, it is possible to reconstruct the trajectory of the user's movements with better accuracy. Other information like the user's hand orientation and the angular twist velocity are collected as well, but they are not needed for the generative model, since they are not used to generate new movements. They are collected to provide a more complete representation of the user's movements, and to eventually allow for a more detailed analysis of the data.

```
... Welcome ... What's your dominant hand? (If no preference is
expressed, pick left hand when possible). (Prepare the environment on the
subject choice).

The device to your (left/right) side is a haptic device, which is connected
to the screen in front of you (the screen shows an empty environment). In
the screen, you can observe the effects that manipulating the device has on
the environment.

Please extract the device from its housing and handle it as if it was a
pen or a precision tool. Feel free to explore the environment in the
simulation. Try to exit from it: you will feel forces that will prevent
you from leaving it. Those are the limits within you can freely move.
There are two buttons on the pen: the darker one lets you rotate the
environment, the lighter one to zoom. [mouse]

(Once the subject learnt how to use it). Now, the screen will show you some
colored spheres (inside other transparent spheres?). Try to make a smooth,
fluid and natural movement that crosses all the spheres. The order in which
the movement must cross the spheres is given by their number.

Initially, when you try to cross the spheres, the device will help you reach
them by pulling you towards their center. After a while, the device will
let you traverse to the next sphere, with a slight pulling force.

After a fixed amount of repetitions, the screen won't show any information,
and only the haptic feedback will be left, which pulling force will decrease
execution by execution.

After a fixed amount of repetitions, the pulling force will decrease to zero
(it must dynamically adjust itself by studying the subject accuracy), from
there you will be in the recording phase.

(Should the user be told when they're in the recording phase?)

Every time you want to be placed to the start position and/or reset the
execution, just press the dark button. Now please, try to cross the spheres
with the aid of the device.

[steps: (sight + haptic feedback), (haptic feedback), (no info)]

(When the pre training phase ends) Congrats. Now, proceed to repeat the
same movement you've done before, without the screen information (or
with only the points visible?) as accurately as possible. If you feel
you're off track, don't fix the movement and try again, since it would be
discarded anyways. (In order to help you, the device will automatically
place itself in the starting position when you press the lighter button. A
brief vibration will tell you when to start the movement. If during the
movement you feel the same vibration, then you have to redo it again?)
```

Those are the same inputs and outputs of the generative model, with the exception that it won't follow the same policies, but only understand the underlying principles from the dataset.

4.4 Experimental Data

Through the executions of the experiments, the system collects a dataset of human movements, expressed as a sequence of timestamped points in the space, with respect to the reference frame.

4.5 Statistical Significance

4.6 System Evaluation

5 Conclusion

Write the conclusion here.

6 Bibliography

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